SEMESTER 2 (2018-19)
Q2 (b)

Q: Consider the system of ODE

$$\frac{dx}{dt} = 2x + y + x^2 - xy$$

$$\frac{dy}{dt} = 3Mx + My - x^2 y$$

Find the value of M for which Hopf-bifurcation occurs at the origin $(x,y)=(0,0)$. How does the nature of equilibrium point change at the Hopf-bifurcation?

Sol^n: Fixed point is $(0,0)$

Linearization gives the Jacobian as

$$J = \begin{bmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial g/\partial x & \partial g/\partial y \end{bmatrix} = \begin{bmatrix} 2+2x-y & 1-x \\ 3M-2xy & M-x^2 \end{bmatrix}$$

$$J_{(0,0)} = \begin{bmatrix} 2 & 1 \\ 3M & M \end{bmatrix} \Rightarrow \begin{array}{l} \tau = 2+M, \quad \Delta = 2M-3M = -M \\ D = \tau^2 - 4\Delta = (M+2)^2 + 4M \end{array}$$

The eigenvalues are: $\lambda_\pm = \dfrac{\tau \pm \sqrt{D}}{2}$

$$\lambda_\pm = \frac{M+2 \pm \sqrt{(M+2)^2 + 4M}}{2}$$

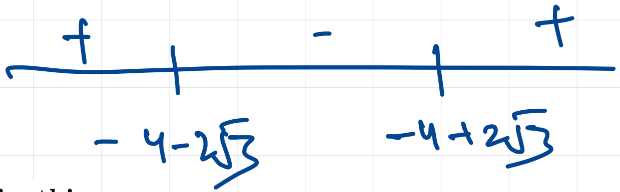
Interesting values of M could be:

$\tau = 0 \Rightarrow M = -2$

$\Delta = 0 \Rightarrow M = 0$

$D = 0 \Rightarrow M^2 + 8M + 4 = 0 \Rightarrow M = \dfrac{-8 \pm \sqrt{64-16}}{2}$

D



$\Rightarrow M = -4 \pm 2\sqrt{3}$

Note: $\lambda_{\pm}\Big|_{\mu=-2} = \pm \frac{1}{2} i 2\sqrt{2} = \pm\sqrt{2}\, i$

and $\frac{d\lambda}{d\mu}\Big|_{\mu=-2} = \left[1 \pm \frac{1}{2}\left[(\mu+2)^2 + 4\mu\right]^{-1/2} \cdot \left(2(\mu+2)+4\right)\right]\frac{1}{2}$

$\neq 0$

From the Hopf-Bifurcation theorem,
we can say that Hopf-Bifurcation occurs at $\mu=-2$.

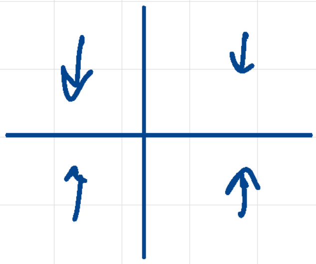
We also know,

for $-4-2\sqrt{3} < \mu < -2$, $(0,0)$ is a stable spiral
since $\tau < 0$ & $D < 0$

for $-2 < \mu < -4+2\sqrt{3}$, $(0,0)$ is an unstable spiral
since $\tau > 0$ & $D < 0$

Asymptotic Analysis at $\mu=-2$:

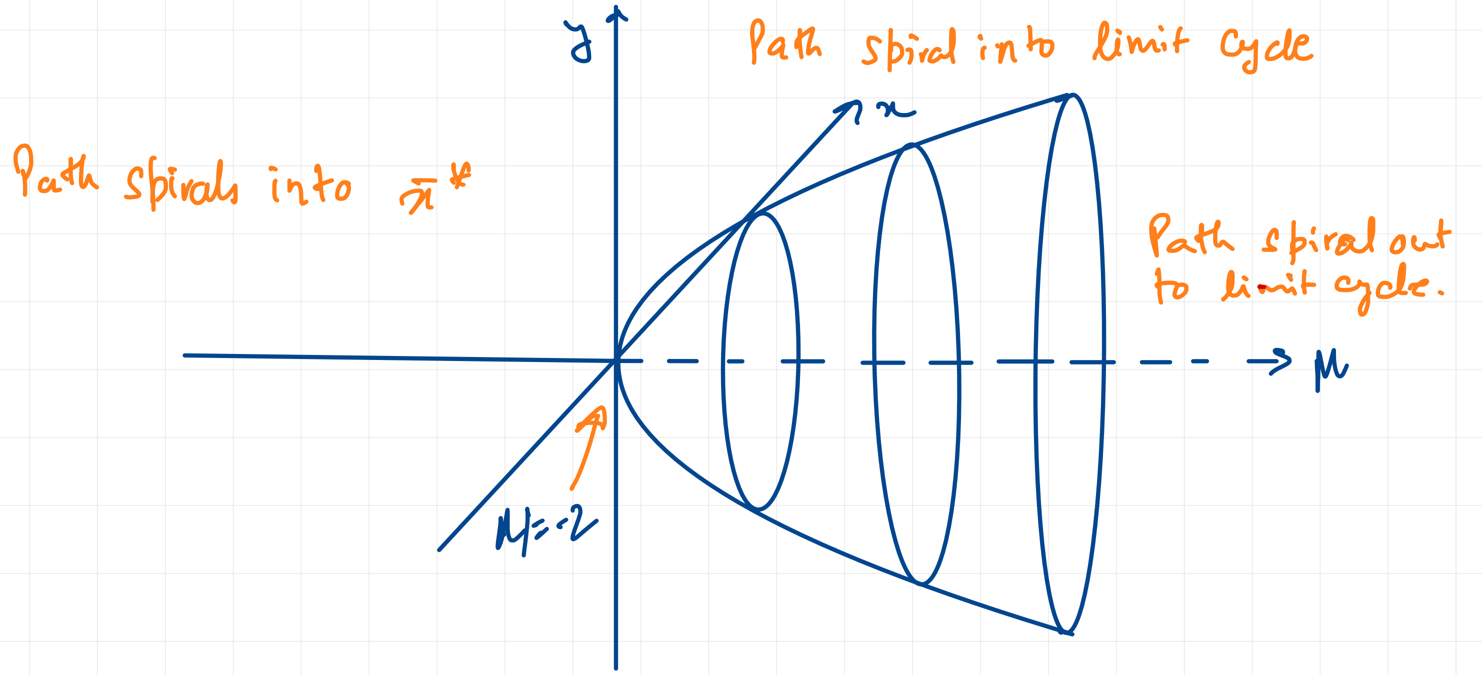
	$\dot{x}$	$\dot{y}$
$x\to\infty, y\to\infty$		$\downarrow$
$x\to\infty, y\to-\infty$		$\uparrow$
$x\to-\infty, y\to\infty$		$\downarrow$
$x\to-\infty, y\to-\infty$		$\uparrow$



	$\mu<-2$	$\mu>-2$
near	Path spirals into $(0,0)$	Path spiral out from $(0,0)$
far	Path move towards $(0,0)$	Path move towards $(0,0)$
	Consistent	Inconsistent

This is an inconsistent situation unless a stable attractor limit cycle occurs in the intermediate region which attracts all the non-equilibrium trajectories in $\mu > -2$

Bifurcation Diagram:



Supercritical Hopf Bifurcation occurs at $\mu = -2$